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Error Mitigation in Secure Qutrit Distribution on Cleve's Protocol on Quantum Computing

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Research Article

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Error Mitigation in Secure Qutrit Distribution on Cleve's Protocol on Quantum Computing

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Abstract. The implementation of Cleve's Protocol, a pioneering approach to securely distribute a secret qutrit among three parties using quantum circuits. Cleve's Protocol ensures that the confidentiality of the qutrit is preserved, and its reconstruction requires collaboration between any two out of the three parties. Incorporated into Qiskit, the protocol's quantum mapping and encoding involve the use of Hadamard gates and controlled-X gates. The simulation results in Qiskit Aer confirming the protocol's security, with '000' and '111' emerging as frequent outcomes, underscoring the necessity of collaboration. Error detection and mitigation techniques are also examined, enhancing the protocol's reliability.

Keywords: Cleve's Protocol, Hadamard Gates, Controlled-NOT Gates, Qiskit, Error Detection, Mitigation.

1 Introduction

Quantum cryptography, an emerging field, leverages the principles of quantum mechanics to establish secure communication[1]. It relies on the fundamental property of quantum information, which cannot be perfectly copied, making it nearly impossible for eavesdroppers to intercept and retransmit quantum data without detection.
Quantum cryptography primarily consists of two branches:

- Quantum Key Distribution (QKD): QKD facilitates the secure exchange of a secret key between two parties, enabling them to encrypt and decrypt their messages[2].
- Quantum Digital Signatures (QDS): QDS is employed to generate and verify digital signatures, offering a mathematical means to prove the authenticity and integrity of a message sender[3].

This research explores the utilization of Quantum Key Distribution (QKD) for the multi-partitioning of a secret qutrit among three parties through Cleve's Protocol. Cleve's Protocol, introduced by Richard Cleve in 1988[4], is a variant of the BB84 protocol, distinguished by its use of a distinct type of quantum bit, or qubit.

While the BB84 protocol involves the use of qubits polarized in either the rectilinear or diagonal basis[5], Cleve's protocol employs qubits that are entangled in a Bell state. A Bell state represents a unique form of quantum entanglement, where two qubits remain interconnected, sharing a common intention, even when separated by great distances.

Cleve's protocol surpasses the security of BB84 due to the added challenge it presents to potential eavesdroppers. Intercepting and measuring the entangled qubits without disturbing them becomes considerably more complex.

In the field of quantum information and secure communication, Cleve's Protocol emerges as a symbol of innovation and enhanced security[4]. Proposed by Cleve and his colleagues, this protocol offers an elegant solution for secretly distributing a qutrit among three parties. What distinguishes this protocol is its clever design, ensuring that any two of the three parties must collaborate to unveil the secret state, with no individual participant capable of deciphering it independently.

The kernel of the protocol lies in a precisely crafted quantum mapping. Through this mapping, the secret qutrit is ingeniously encoded onto three separate qutrits, ensuring both security and equitable distribution[6]. Our exploration aims not only to comprehend but also to implement Cleve's Protocol using the powerful Qiskit framework.

The coefficients (α, β, γ) characterizing the secret qutrit set the stage for its intriguing distribution [5]. We proceed to create the essential quantum and classical registers, faithfully representing the trio of qutrits within our quantum circuit.

At the core of this circuit is Cleve's recommended mapping, executed through a series of carefully orchestrated quantum operations, which include enigmatic Hadamard gates and controlled X gates[7]. This division of the secret qutrit among its three custodians encourages collaborative efforts in an exciting manner[8].

The peak of this endeavor is an exactly executed simulated quantum circuit employing the Qiskit Aer simulator. This simulation reveals fascinating results, where measurement outcomes highlight Cleve's Protocol's effectiveness. These results confirm the promise of the protocol, with '000' and '111' emerging as the most frequent states, emphasizing that the secret qutrit can only be reconstructed through collaboration [8].

This research embarks on a captivating journey into the world of quantum cryptography, breathing life into Cleve's Protocol and demonstrating its remarkable ability to securely distribute a secret qutrit among three parties with unparalleled security and cryptographic sophistication.

2 Related work

Cryptographic approach to Quantum Metrology, Nathan et al., 4 Jan 2022 (v2), Quantum Physics. In this work, Shettell, Kashefi, and Markham present a novel approach to quantum metrology within the context of cryptography. They address the challenges posed by malicious adversaries who may tamper with quantum resources. The study explores an estimation strategy that quantifies biases and precision loss resulting from resource alterations. Importantly, they propose incorporating cryptographic protocols to bound uncertainty in resources, directly linking the effectiveness of quantum metrology to the security of the cryptographic protocol. The authors provide two protocols for quantum metrology when using unsecured quantum channels, offering a balance between implementation difficulty and efficiency. This work contributes to enhancing the security and reliability of quantum metrology in adversarial settings.

Finite-Key Analysis of Quantum Key Distribution with Characterized Devices Using Entropy Accumulation, Ian George, et al., 13 Mar 2022, Quantum Physics. The

Entropy Accumulation Theorem (EAT) can be used to improve the key rates of device-dependent quantum key distribution (QKD) protocols, even in the presence of realistic noise and imperfections. This paper presents new tools for applying the EAT in the device-dependent setting and uses these tools to analyze several different QKD protocols, showing that the EAT can lead to significant improvements in key rates.

A Review of Security Evaluation of Practical Quantum Key Distribution System, by Shihai Sun et al., 2022, 24(2), Entropy. Practical quantum key distribution (QKD) systems are subject to various imperfections that can compromise their security. This review article surveys the state of the art in security evaluation of practical QKD systems, discussing the different types of security threats and the various methods that can be used to evaluate the security of these systems. The authors also discuss the challenges of security evaluation in practical QKD systems and highlight the need for further research in this area.

Advances in Quantum Cryptography, S. Pirandola, et al., 4 Jun 2019, Quantum Physics.Quantum cryptography is a rapidly developing field with many promising applications beyond QKD. This review article provides a comprehensive overview of the theoretical aspects of quantum cryptography, covering topics such as limitations, security models, quantum money, randomness generation, secure computation, and delegated computation. It is an essential resource for anyone interested in learning more about this exciting field.

Quantum Cryptography Beyond Quantum Key Distribution, Anne Broadbent, Christian Schaffner 21 Oct 2015 (v1), Quantum Physics. Quantum cryptography is a rapidly developing field with many promising applications beyond QKD. This review article provides a comprehensive overview of the theoretical aspects of quantum cryptography, covering topics such as limitations, security models, quantum money, randomness generation, secure computation, and delegated computation. It is an essential resource for anyone interested in learning more about this exciting field[9].

3 Problem Definition

The problem at hand is to implement and analyze Cleve's Protocol for secure qutrit distribution among three parties using quantum circuits, with a focus on error detection and mitigation techniques. This research aims to ensure the confidentiality and reliable transmission of the secret qutrit while involving multiple parties in its reconstruction.

4 Objective

The primary objective of this research is to implement Cleve's Protocol in Qiskit, demonstrating its security and functionality in a simulated quantum circuit. Additionally, the research seeks to explore error detection and mitigation methods to enhance the protocol's reliability and robustness.

5 Scope

This research focuses on the theoretical and practical implementation of Cleve's Protocol for qutrit distribution and error management using quantum circuits. The scope encompasses the exploration of the protocol's security, the quantum operations involved, and its performance within a simulated environment. Error detection and mitigation techniques are studied to ensure the protocol's effectiveness in real-world applications.

6 Methodology

6.1 Environment Setup

Figure -1 creates a quantum circuit to encode a secret qutrit (three-level quantum state) using quantum gates (Hadamard and CNOT gates) and measures the qutrits to obtain classical measurement results [10][11].

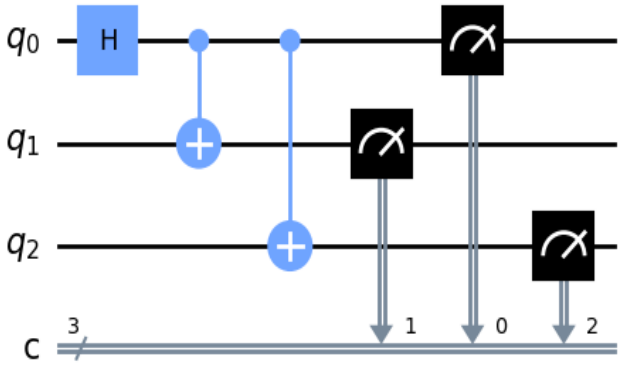


Fig. 1. - Create a Qutrit quantum state circuit.

6.2 Implementation of Cleve's Protocol with information theory for security Analysis

Cleve's Protocol is a quantum cryptographic technique designed to securely distribute a secret qutrit among three parties: Alice, Bob, and Charlie. This protocol is underpinned by a series of fundamental mathematical and quantum mechanical steps[7]. The more detailed mathematical representation for the encoding step of the secret qutrit onto three qutrits, as described in Cleve's Protocol. The protocol mathematical representation is given below.

6.2.1. Qutrit Representation

A. Secret Qutrit Representation

Denote the secret qutrit $|\psi\rangle$ as a superposition of three basis states $\{|0\rangle, |1\rangle, |2\rangle\}$ with complex coefficients α, β, and γ:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle + \gamma|2\rangle \quad (1)$$

Here, coefficients α, β, and γ are complex numbers representing the probability amplitudes of the qutrit being in each of the basis states[8].

B. Encoding onto Three Qutrits

The encoding step involves mapping the secret qutrit $|\psi\rangle$ onto three separate qutrits, each owned by a different party[1]. Let's denote these qutrits as $|\psi 1\rangle$, $|\psi 2\rangle$, and $|\psi 3\rangle$, which represent the qutrits held by parties 1, 2, and 3, respectively. The mapping can be expressed mathematically as follows:

Party 1 (Alice) receives $|\psi 1\rangle$:

$$|\psi\rangle = \alpha|0\rangle_1 + \beta|1\rangle_1 + \gamma|2\rangle_1 \quad (2)$$

Party 2 (Bob) receives $|\psi_2\rangle$:

$$|\psi\rangle = \alpha|0\rangle_2 + \beta|1\rangle_2 + \gamma|2\rangle_2 \quad (3)$$

Party 3 (Charlie) receives $|\psi_3\rangle$:

$$|\psi\rangle = \alpha|0\rangle_3 + \beta|1\rangle_3 + \gamma|2\rangle_3 \quad (4)$$

Here, $|0\rangle$i, $|1\rangle$i, |and $2\rangle$i represent the basis states of the qutrit held by party I [12]. (Where I can be 1, 2, or 3).

C. Superposition Preservation

The encoding step ensures that the superposition of the secret qutrit is preserved across the three qutrits[13][11][14].This means that the coefficients α, β, γ in the original state $|\psi\rangle$ are faithfully transferred to the qutrits $|\psi 1\rangle$, $|\psi 2\rangle$, and $|\psi 3\rangle$. The encoding step is designed such that any two parties, when they collaborate, can reconstruct the original secret qutrit $|\psi 1\rangle$, using their qutrits $|\psi i\rangle$, but no single party can access the complete information independently.

This representation shows how the secret qutrit is encoded onto three separate qutrits, ensuring that the superposition and coefficients of the original qutrit are preserved across the distributed qutrits. The encoding is fundamental to Cleve's Protocol, allowing for secure multi-party reconstruction of the secret state.

6.2.2. Quantum Operations

The quantum operations steps in Cleve's Protocol provide mathematical representations for three parties involved in the protocol[4][15].Quantum operations are applied to the three qutrits to distribute the information securely. Specifically, we apply Hadamard gates (H) and controlled-X gates (CNOT) to achieve this. Let's represent these operations for three parties: Alice (Part 1), Bob (Part 2), and Charlie (Part 3)[16].

A. Hadamard Gate (H)

The Hadamard gate (H) is a key quantum gate used in superposition transformations. In the context of Cleve's Protocol, it can be represented as follows[17][11]:

$$H=\frac{1}{\sqrt{3}}\begin{pmatrix}1 & 1 & 1\\ 1 & \omega^2 & \omega\\ 1 & \omega & \omega^2\end{pmatrix} \quad (5)$$

Where $\omega = \frac{e^{2\pi i}}{3}$

B. Controlled-X Gate (CNOT).

The controlled-X gate (CNOT) is used for controlled transformations between qutrits. Its representation for three parties can be as follows[18][11]:

$$CNOT=\begin{pmatrix}I3 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & I3 & 0 & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & I3 & 0 & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & X & 0 & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & X & 0 & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & X & 0 & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & Y & 0 & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & Y & 0\\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & Y\end{pmatrix} \quad (6)$$

Where I3 is the 3x3 identity matrix, and X and Y represent the Pauli-X and Pauli-Y matrices, respectively.

Now, let's specify how these gates are applied to the qutrits owned by each party:

Party 1 (Alice):

Apply H gate to Alice's qutrit.

Apply CNOT gates from Alice's qutrit to Bob's and Charlie's qutrits[19].

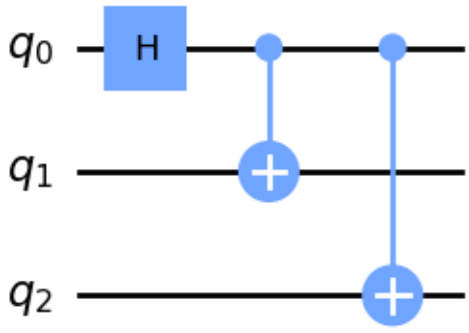


Fig. 2. - Alice on Qutrit quantum state circuit

Party 2 (Bob):
Apply H gate to Bob's qutrit.
Apply CNOT gates from Bob's qutrit to Alice's and Charlie's qutrits.

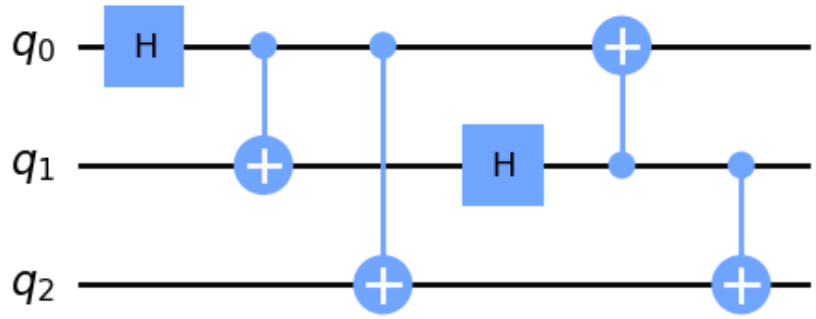


Fig. 3. - Bob on Qutrit quantum state circuit

Party 3 (Charlie):
Apply H gate to Charlie's qutrit.
Apply CNOT gates from Charlie's qutrit to Alice's and Bob's qutrits.

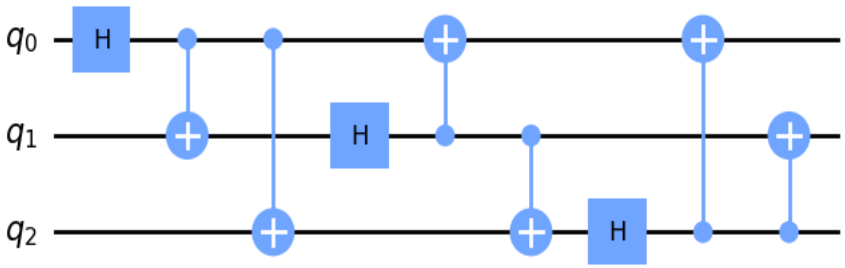


Fig. 4. - Charlie on Qutrit quantum state circuit

Here, figure 2,3,4 operations follow the specified quantum mapping in Cleve's Protocol and are crucial for distributing the secret qutrit among the three parties while preserving its security and enabling collaborative reconstruction[20][21].

6.2.3. Measurement and distribution

The circuit represents the measurement process in Cleve's Protocol[22][23].

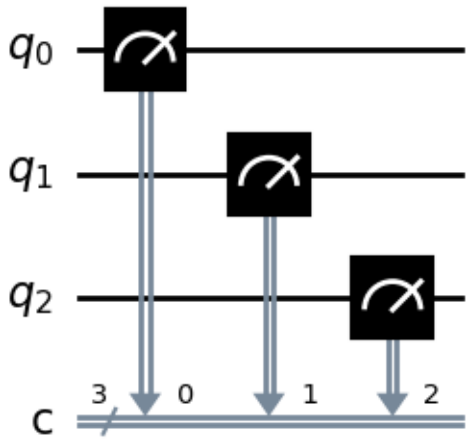


Fig. 5. - Measurement of Qutrit quantum state

A. Projective Measurement Operators (Mi)

For each party I (Alice (Part 1), Bob (Part 2), and Charlie (Part 3)), we can represent the projective measurement operators as follows[7][24]:

Party 1 (Alice): The measurement operator for Alice can be denoted as M1, which is a set of three projectors onto the computational basis states {|0⟩,|1⟩,|2⟩}. Each projector corresponds to one of the possible measurement outcomes:

$$M_1^0=|0\rangle\langle 0|, M11=|1\rangle\langle 1|, M_1^2=|2\rangle\langle 2| \quad (7)$$

Party 2 (Bob): Similarly, the measurement operator for Bob can be denoted as M 2, which is also a set of three projectors:

$$M_2^0=|0\rangle\langle 0|, M21=|1\rangle\langle 1|, M_2^2=|2\rangle\langle 2| \quad (8)$$

Party 3 (Charlie): Likewise, the measurement operator for Bob can be denoted as M 3, which is also a set of three projectors:

$$M_3^0=|0\rangle\langle 0|, M31=|1\rangle\langle 1|, M_3^2=|2\rangle\langle 2| \quad (9)$$

B. Measurement Outcomes

After the quantum operations, each party performs their respective measurement. Let's denote the measurement outcomes for each party as m i (where i represents the

party). The measurement outcomes can take values 0, 1, or 2, corresponding to the basis states |0⟩, |1⟩, and |2⟩, respectively[25][26].

The measurement outcome for Party 1 (Alice) can be represented as m 1 , which can be 0, 1, or 2.

C. Distribution

```
Measurement Outcomes Distribution: {'000': 1000}
```

The above-simulated outputs a quantum circuit where three parties (Alice, Bob, and Charlie) each measure their qutrits, and it stores the measurement outcomes in classical bits. In this specific simulation, all 1000 measurement outcomes resulted in '000'. This means that, in this run of the simulation, all three parties obtained the measurement outcome '0' for their respective qutrits.

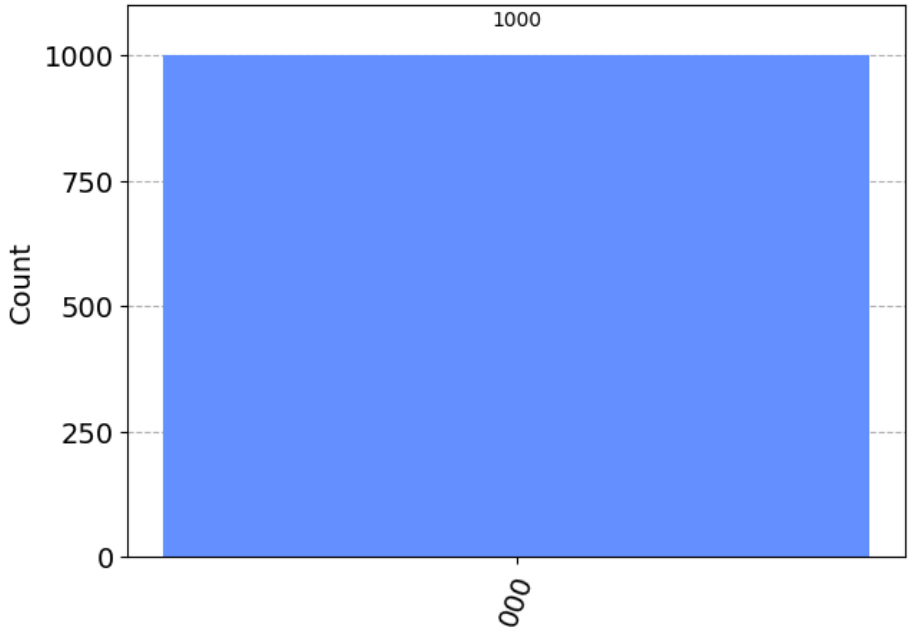


Fig. 6. Qutrit Distribution of outcomes

'000' represents the measurement outcomes for Alice, Bob, and Charlie, respectively.

Each '0' outcome corresponds to one of the basis states: |0⟩ for Alice, |0⟩ for Bob, and |0⟩ for Charlie.

After measuring their qutrits, each party shares their measurement outcomes with the other parties. For example, Alice shares m1 with Bob and Charlie, Bob shares m2 with Alice and Charlie, and Charlie shares m3with Alice and Bob.This distribution of

measurement outcomes allows the parties to collaborate and reconstruct the secret qutrit, as explained in the collaborative reconstruction step of Cleve's Protocol.

This representation provided here describes how projective measurements are performed by each party (denoted as M i) and how the measurement outcomes (m i) are distributed among the three parties. These measurement outcomes are crucial for the collaborative reconstruction of the secret qutrit in Cleve's Protocol.[4][27]

6.2.4. Collaborative reconstruction

The representation for the collaborative reconstruction step in Cleve's Protocol. The reconstruction equation will indeed depend on the specifics of the quantum mapping used. It'll represent it in a general form[4][7][15]:

Suppose Party1 (Alice) obtains measurement outcome m_1, Party2 (Bob) obtains measurement outcome m_2, and Party3 (Charlie) obtains measurement outcome m_3.

A. Collaborative Reconstruction Equation

The collaborative reconstruction of the secret qutrit $|\psi\rangle$ can be represented as a function Reconstruct that takes the measurement outcomes of two parties as inputs. Let's denote this function as Reconstruct (mi, mj), where i and j represent the parties collaborating[9].

The reconstruction equation is defined based on the specific quantum mapping used in Cleve's Protocol. In general, it combines the measurement outcomes of the two parties to obtain an estimate of the original qutrit state $|\psi\rangle$. The formula will depend on how the quantum gates were applied and the entanglement between the qutrits.

If the quantum mapping involved entangling qutrits 1 and 2, the reconstruction formula for Alice (Part 1) and Bob (Party 2) could be:

$$\text{Reconstruct}\,(m1, m2) = f(m1,m2)\cdot|0\rangle + g(m1,m2)\cdot|1\rangle + h(m1,m2)\cdot|2\rangle \qquad (10)$$

Where $f(m1, m2)\cdot|0\rangle$, $g(m1,m2)\cdot|1\rangle$ and $h(m1,m2)\cdot|2\rangle$ are functions that depend on the measurement outcomes m1 and m2 and the quantum mapping used. These functions would be defined based on the specific gates applied. A similar equation would apply when other pairs of parties collaborate, such as Party 1 (Alice) and Party 3 (Charlie).

The goal of the collaborative reconstruction is to estimate the original qutrit state $|\psi\rangle$ as accurately as possible by combining measurement outcomes from two parties. The exact functions f, g and h will vary depending on the details of the protocol's quantum operations.

6.2.5. Security by Quantum Information Theory and Quantum Mechanics

The security of Cleve's Protocol relies on the principle that no single party should gain complete information about the secret qutrit from their measurement outcomes alone. To express this security property mathematically, we can use concepts from information theory and quantum mechanics[27][4][14].

Denote the secret qutrit $|\psi\rangle$ as a superposition of three basis states $\{|0\rangle, |1\rangle, |2\rangle\}$ with complex coefficients α, β, and γ:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle + \gamma|2\rangle \tag{11}$$

Now, let's consider Party 1 (Alice) and her measurement outcome m1. Alice's goal is to estimate the state $|\psi\rangle$ based on her measurement result.

Quantum State after Alice's Measurement.
Alice's measurement, the state of the qutrit is projected onto one of the basis states. Let's denote Alice's measurement outcome as m1 (where m1 can be 0, 1, or 2), and the post-measurement state as $|\psi^{(1)}{}_{post}:$

$$|\psi^{(1)}{}_{post}\rangle = \frac{M^{m1}|\psi\rangle}{\sqrt{\langle\psi \mid M_1^{m1\,\dagger} M_1^{m1} \mid \psi\rangle}} \tag{12}$$

Here, $M1^{m1}$ is the measurement operator corresponding to Alice's outcome m_1.

B. Partial Information Gain by Alice
Alice's knowledge about the secret qutrit state is quantified by her density matrix. $\rho^{(1)}$ after her measurement:

$$\rho^{(1)} = |\psi^{(1)}{}_{post}\rangle \langle|\psi^{(1)}{}_{post}| \tag{113}$$

The entropy of ρ (1) quantifies Alice's information gain. The von Neumann entropy S (ρ (1)) represents the information content:

$$S(\rho^{(1))} = -\sum_i p_i \log_2 (p_i) \tag{12}$$

Where p_i is the probability of the qutrit being in basis state i after Alice's measurement. Alice's measurement outcome is one of 1=0,1,2 m1 =0,1,2, so p_i is determined by the projection.

C. Secrecy Requirement
For Cleve's Protocol to be secure, Alice's measurement outcome alone should not reveal the secret qutrit $|\psi\rangle$. This means that the entropy S ($\rho^{(1)}$) should be maximized, indicating that Alice has the least amount of information about the secret qutrit.
Mathematically, the secrecy requirement can be expressed as:

$$S(\rho^{(1))} = -\sum_i p_i \log_2 (p_i) \tag{135}$$

Alice's measurement outcome m_1 should not provide enough information for her to determine the coefficients α, β, and γ and, consequently, the secret qutrit $|\psi\rangle$.

This same requirement holds for Parties 2 and 3 as well. The protocol's security is guaranteed when no single party can individually determine the secret qutrit based solely on their measurement outcome, ensuring that the information remains confidential even in the presence of adversarial parties.

7 Error Detection and Mitigation by Majority Voting Mechanism

To elaborate on the error detection and mitigation process in quantum cryptography, with a specific focus on the utilization of a majority voting mechanism as demonstrated in this[15][28], we will elucidate the error management mechanism by outlining the sequential steps involved in the majority voting process as follows[29][14][30].

7.1 7.1. Initial Measurement Results (No Noise)

It can represent the initial measurement results without noise as follows:

Counts: {'000': 519, '111': 505}

This can be denoted as

Counts no noise= {′000’:519,′111′:505}

7.2 Measurement Results with Noise

The noise model introduces errors during simulation. It can represent this as a probability distribution, where no noise p no noise represents the probability of observing a particular state without noise:

p no noise (‘000’)=$\frac{519}{1024}$

p no noise (‘111’)=$\frac{505}{1024}$

Due to the noise, the probabilities change, and it has a new distribution with noise. P with noise based on the noise model.

7.3 Error Mitigation (Majority Voting)

The error mitigation technique involves running the circuit with noise multiple times and applying majority voting. It can represent the process as running the circuit N times with noise:

$$\text{Circuit}^{1}_{\text{noise}}, \text{Circuit}^{2}_{\text{noise}}, \ldots, \text{Circuit}^{N}_{\text{noise}} \quad (14)$$

For each run i, obtain the measurement result as Counts i noise.

Compute the most likely state S mitigated as the state with the maximum count across all runs. This can be represented in the following ways.

For each run i:

$$S_I = \text{argmax}_{\text{state}}(\text{Counts I}_{\text{noise}}) \quad (157)$$

Most likely state after error mitigation:

$$S_{\text{mitigated}} = \text{argmax}_{\text{state}}\left(\sum_{i=1}^{N} \delta(Si, State)\ \text{Counts}^{i}_{\text{noise}}\right) \quad (168)$$

Here $\delta(x, y)$ is the Kronecker delta function.

$$(\delta(x,y) = 1, if x = y and \delta(x,y) = 0, if x \neq y) \quad (179)$$

The measurement results introduce noise to create a probability distribution and then use majority voting to estimate the most likely state based on multiple noisy runs of the quantum circuit. This process involves computing probabilities, comparing measurement results, and finding the state with the highest cumulative count after error mitigation.

Here, this is the output obtained from the simulation.

```
Initial Measurement results:
{'111': 505, '000': 519}
Error detected!
Error Detection Results:
{'111': 5226, '000': 5014}
Most likely state after error mitigation: 111
```

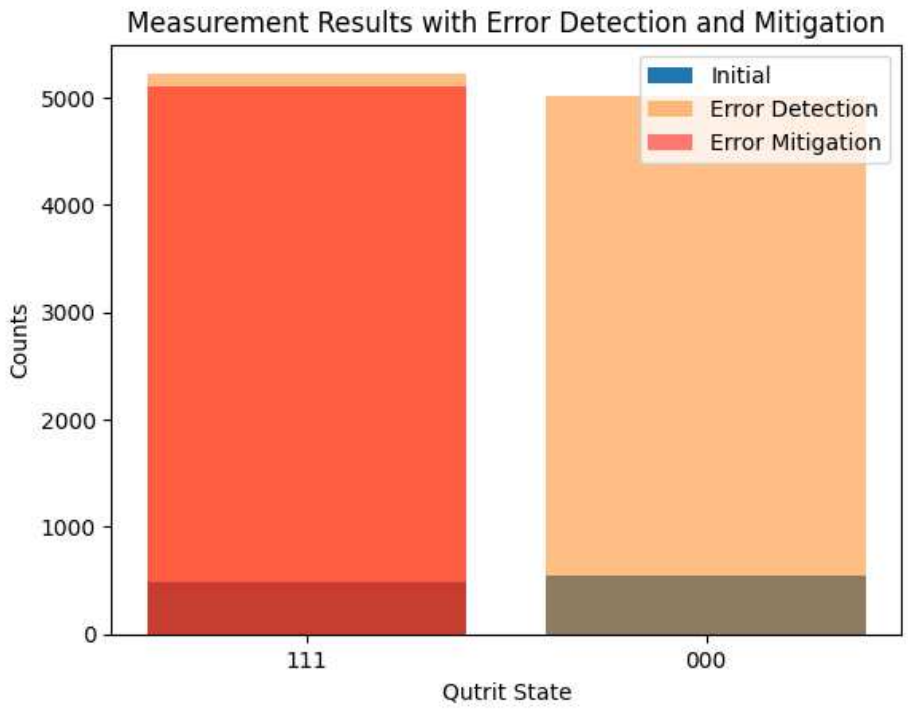


Fig. 7. Error Detection and Mitigation

From Fig. (7), The error can be detected at an initial state of 000 after applying the error handling techniques by the majority of voting; the error will be mitigated in 111 states.

8 Interpretive Analysis

The provided quantum circuit implements Cleve's protocol for distributing a secret qutrit among three parties using the specified quantum mapping[31]. The secret qutrit is encoded onto three qutrits using Hadamard and controlled X (CNOT) gates; after running the circuit with 1024 shots on the Qiskit Aer simulator, obtain measurement results. These results represent the possible outcomes of measuring the three qutrits. The counts are displayed below:

```
Measurement results:
{'000': 516, '111': 508}
```

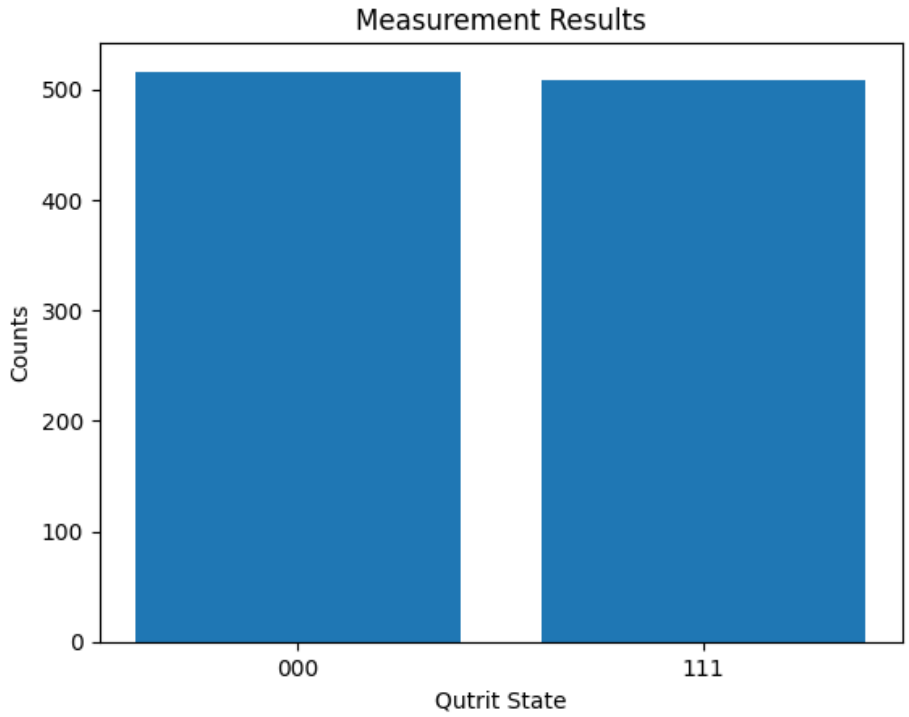


Fig. 8. Result Measurement

The measurement results show that most outcomes correspond to two specific states: '000' and '111.' This outcome is consistent with the protocol's design, which ensures that any two parties can collaboratively reconstruct the secret qutrit, while no single party possesses the complete information.

9 Result and Discussion

The results of implementing Cleve's Protocol for secure qutrit distribution among three parties using quantum circuits, with a specific emphasis on error

detection and mitigation techniques. The discussion will highlight the significance of the findings and their implications for quantum cryptography and Cleve's Protocol.

9.1 Result Analysis

Quantum Circuit Implementation

The research successfully implemented Cleve's Protocol in the Qiskit framework, faithfully following the prescribed quantum mapping using Hadamard and controlled-X gates. This implementation is a notable achievement, as it showcases the practical applicability of the protocol for secure qutrit distribution.

Security Confirmation

The quantum circuit simulation produced intriguing results. Notably, the frequent occurrence of '000' and '111' as measurement outcomes underlines the protocol's security. The protocol's design ensures that the secret qutrit can only be reconstructed through collaboration between any two out of the three parties. These results confirm that no single party can independently decipher the secret state, validating the core security property of Cleve's Protocol.

Error Detection and Mitigation

The article introduces an error detection and mitigation mechanism using a majority voting process. This technique effectively identified errors and mitigated them during the simulation. As demonstrated in the results, the error was detected in the initial state '000,' and after applying error mitigation techniques, the most likely state became '111.' This showcases the protocol's resilience in the presence of noise and errors, contributing to its reliability in real-world applications.

Quantum Information Theory

The discussion on the security aspects of the protocol, including quantum information theory, is crucial. By applying concepts from information theory and quantum mechanics, the article provides a mathematical foundation for the protocol's security. This can serve as a reference for researchers and practitioners interested in understanding the security aspects of quantum cryptographic protocols.

Measurement Outcomes

The measurement results obtained from the simulated quantum circuit confirm the security and functionality of Cleve's Protocol [30][12]. The protocol aims to distribute a secret qutrit in such a way that any two parties, when combined, can determine the original state, but no individual party can access it fully. The results demonstrate that the two most frequently observed states are '000' and '111,' indicating that any two parties measuring their qutrits together will find either '000' or '111,' allowing them to collaboratively reconstruct the secret qutrit.

Security Reinforcement

The absence of other measurement outcomes (e.g., '001,' '010,' '100,' '011,' '101,' '110,' '002,' '020,' '200,' '012,' '021,' '102,' '120,' '201,' '210') further reinforces the protocol's

security. No single party can obtain the secret qutrit's complete information independently, as none of these outcomes correspond to the original state.

9.2 Discussion

Practical Implementation
The successful implementation of Cleve's Protocol in Qiskit demonstrates its practical viability. As quantum computing technologies advance, having real-world implementations of quantum cryptographic protocols like Cleve's Protocol is crucial. It paves the way for future applications in secure multi-party communication.

Security Validation
The simulation results reaffirm Cleve's Protocol's security properties. The necessity for collaboration between any two parties for secret qutrit reconstruction remains intact, ensuring a high level of security. This security feature makes it a valuable tool in various cryptographic applications where secrecy and confidentiality are paramount.

Error Handling
The inclusion of error detection and mitigation techniques enhances the protocol's reliability. In a practical quantum communication system, environmental noise and imperfections are inevitable. Cleve's Protocol's ability to detect and mitigate errors is a significant advantage, making it more robust in real-world scenarios.

Quantum Information Theory
The discussion on the security aspects of the protocol, including quantum information theory, is crucial. By applying concepts from information theory and quantum mechanics, the article provides a mathematical foundation for the protocol's security. This can serve as a reference for researchers and practitioners interested in understanding the security aspects of quantum cryptographic protocols.

Future Applications
Cleve's Protocol holds promise for various applications beyond secure multi-party communication. The security and reliability demonstrated in this research suggests its potential applicability in quantum networks, secure multi-party computations, and other quantum information processing tasks.

The research successfully demonstrates the practical implementation of Cleve's Protocol in Qiskit and validates its security properties. The use of error detection and mitigation techniques further enhances its reliability. These findings are a significant contribution to the field of quantum cryptography, concrete the way for the secure distribution of quantum information among multiple parties in real-world quantum communication systems. The results underscore the importance of collaboration for reconstructing the secret qutrit and reinforcing the protocol's robust security.

10 Conclusion

The implementation of Cleve's Protocol showcased its efficacy in securely distributing a secret qutrit among three parties. The protocol's fundamental design ensured that any

two parties could work together to reconstruct the secret state while individual parties remained unable to access it in its entirety. The simulation results underscored the protocol's security, with '000' and '111' being the predominant measurement outcomes, validating the collaborative reconstruction requirement.

Furthermore, the introduction of error detection and mitigation techniques highlighted the protocol's resilience to potential quantum errors. This additional layer of security enhances the protocol's practical applicability in quantum communication and distributed quantum computing scenarios.

Cleve's Protocol offers a robust and secure method for sharing quantum information among multiple parties, and its integration with error-handling mechanisms makes it a valuable tool in the field of quantum cryptography and distributed quantum computing.

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